Syrian Private university
Faculty of Artificial Intelligence Engineering
Department of Intelligent Information
Security Systems Engineering

الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

Secure Communication Protocol for Unmanned Aerial Vehicles (UAVs)





Prepared by

Massa Bakleh Zoher Halaweh

Omar Afra

Supervised by

Dr. Wassim Aljuneidi
Eng. Yamen AL Halak

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❑ Problem Statement:

The MAVLink protocol used in UAV communication lacks built-in security mechanisms such as encryption and authentication, making UAV systems vulnerable to attacks including eavesdropping, spoofing, replay attacks, and unauthorized access over wireless networks.



❑ Objectives Statement :

Protect UAV-to-GCS and UAV-to-UAV communications against cyberattacks, including replay attacks.

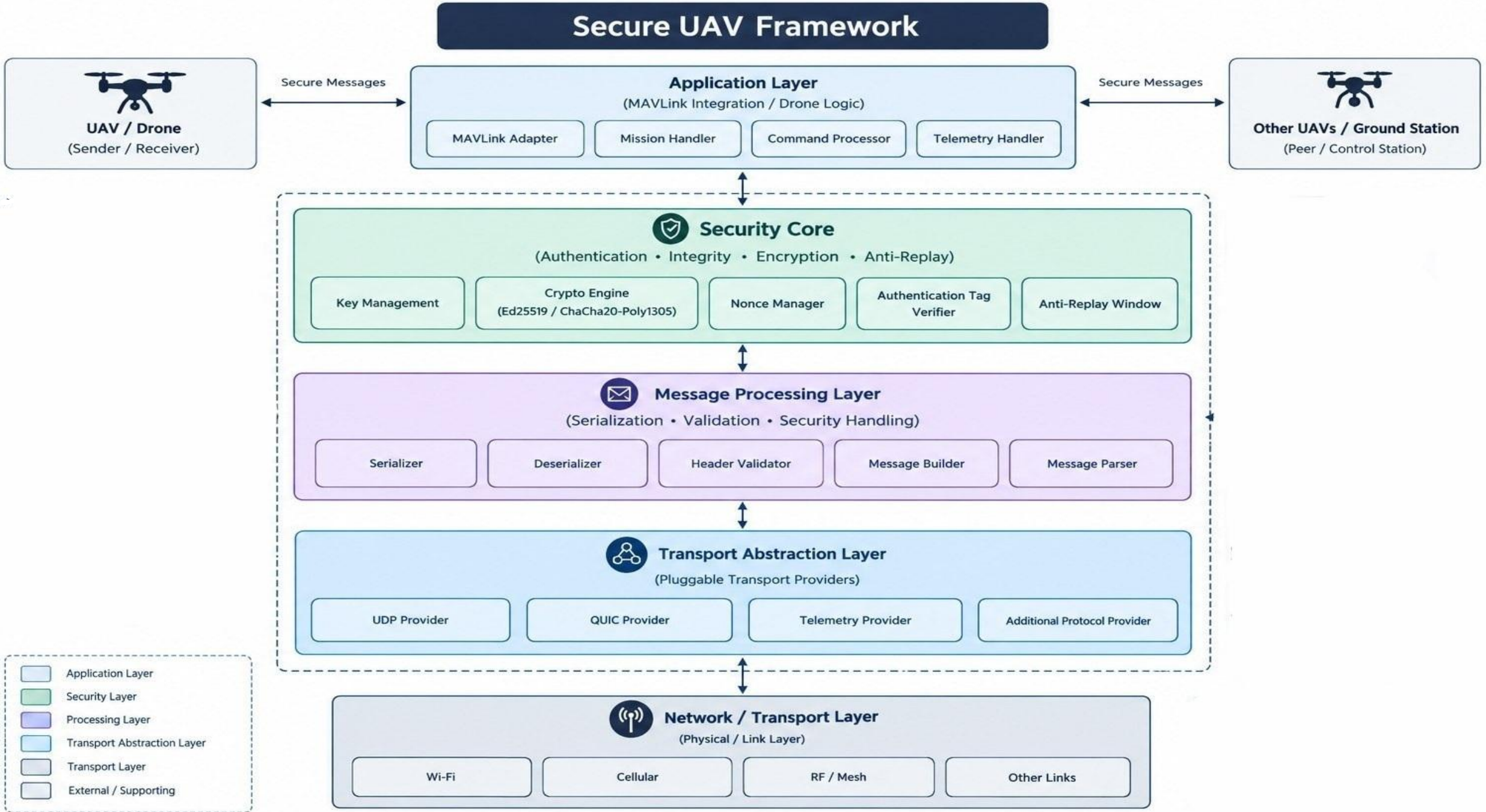
Design and implement a lightweight, modular security framework independent of PX4.

Implement authentication, encryption, integrity verification, and anti-replay protection using ChaCha20-Poly1305.

Integrate the framework with MAVLink 2 and evaluate its resistance to security attacks.

Simulate the framework using NS-3 and evaluate performance metrics, including latency, CPU usage, memory consumption, and packet loss.

Article Title	Publication Year	Journal / Source	Authors	What Was Achieved in This Paper	Reported Performance Metrics	Strengths	Relevance to our Project
MAVLink Protocol: A Survey of Security Threats and Countermeasures	2024	IEEE ICoDT2	Various Authors	Presented a comprehensive survey of MAVLink security threats and countermeasures, covering MAVLink 1/2, confidentiality, integrity, availability, authentication, encryption and key management.	Threat classification; security mechanisms; protocol and key-management analysis	Broad and systematic coverage of MAVLink security	Provides the main background for identifying MAVLink-related threats and deriving V1 security requirements, without requiring the framework to depend on MAVLink.
Provably Secure Authentication Protocol for Traffic Exchanges in UAVs	2024	Research Paper	Various Authors	Proposed a lightweight authentication protocol providing mutual authentication, session-key agreement, anonymity and unlinkability, with formal security analysis.	Computation cost; communication overhead; security/provability analysis	Addresses replay, impersonation, MITM and other attacks while supporting session-key security properties	Supports defining attacker capabilities and analyzing authentication, replay, impersonation, and session-key security requirements.
A Lightweight Authentication Protocol for UAVs Based on ECC Scheme	2023	Drones (MDPI), 7, 315	Zhang; Liu; Han; Yang	Proposed LAPEC, an ECC-based identity authentication and key-agreement protocol for UAVs, including backward secrecy of session keys.	Authentication and key-agreement computation; communication overhead; security analysis	Uses ECC to balance security and resource efficiency and improves session-key security	Provides a comparison point for asymmetric authentication and session-key protection; it does not require changing the selected V1 design.
SLAKA-IoD: A Secure and Lightweight Authentication and Key Agreement Protocol for Internet of Drones	2024	Drones (MDPI), 8, 374	Yuelei Xiao; Yu Tao	Proposed SLAKA-IoD, a lightweight authentication and key-agreement protocol using PUF, XOR, and hash functions. It supports mutual authentication and secure session-key establishment between drone–GS and drone–drone pairs, with informal and formal security analysis and Scyther verification.	Computation and communication cost; security-property analysis; comparison with related IoD authentication/key-agreement schemes	Supports D2G and D2D mutual authentication, session keys, replay resistance, and lightweight cryptographic operations for resource-constrained drones.	Supports the Security Core's authentication and key-management requirements, including mutual authentication, session-key establishment, and replay resistance for D2G/D2D communication.
Lightweight Drone-to-Ground Station and Drone-to-Drone Authentication Scheme for Internet of Drones	2025	Symmetry (MDPI), 17, 556	Qi Xie; Jilin Zhang	Proposed a lightweight D2G and D2D authentication and key-agreement protocol using PUFs. It enables direct mutual authentication between drones without GS involvement and addresses capture, impersonation, privacy, and forward-secrecy concerns.	Average computational-efficiency improvement of 39.44% versus comparable schemes; storage overhead during GS authentication; formal security analysis	Direct D2D authentication without GS, PUF-based protection against capture attacks, perfect forward secrecy, and improved efficiency.	Directly supports the planned UAV–GCS and UAV–UAV use cases, especially mutual D2D authentication without requiring the Ground Station to participate in every exchange.


Secure UAV Framework
UAV / Drone
(Sender / Receiver)
Secure Messages
Application Layer
(MAVLink Integration / Drone Logic)
MAVLink Adapter
Mission Handler
Command Processor
Telemetry Handler
Secure Messages
Other UAVs / Ground Station
(Peer / Control Station)
Security Core
(Authentication • Integrity • Encryption • Anti-Replay)
Key Management
Crypto Engine
(Ed25519 / ChaCha20-Poly1305)
Nonce Manager
Authentication Tag Verifier
Anti-Replay Window
Message Processing Layer
(Serialization • Validation • Security Handling)
Serializer
Deserializer
Header Validator
Message Builder
Message Parser
Transport Abstraction Layer
(Pluggable Transport Providers)
UDP Provider
QUIC Provider
Telemetry Provider
Additional Protocol Provider
Network / Transport Layer
(Physical / Link Layer)
Wi-Fi
Cellular
RF / Mesh
Other Links
Application Layer
Security Layer
Processing Layer
Transport Abstraction Layer
Transport Layer
External / Supporting



❑ System Implementation Phases:





❑ Implementation Environment:

- ns-3 Network Simulator (v3.47)
- Ubuntu Linux 24.04 for Development environment.
- Visual Studio Code .
- C++ Programming Language.
- OpenSSL Library for ChaCha20-Poly1305 encryption.
- GoogleTest for Unit testing.
- MAVLink C Library
- PX4 SITL for UAV simulation.
- Net Anim for visualization.
- Wireshark for packet analysis.



❑ Implemented Phases:

✓ Install px4 platform